

(2½ Hours)

[Total Marks: 75]

- N. B.: (1) All questions are compulsory.
 (2) Make suitable assumptions wherever necessary and state the assumptions made.
 (3) Answers to the same question must be written together.
 (4) Numbers to the right indicate marks.
 (5) Draw neat labeled diagrams wherever necessary.
 (6) Use of Non-programmable calculators is allowed.

1. Attempt any three of the following: 15
 - a. Explain Artificial Intelligence with Turing Test approach.
 - b. Describe the contribution of Philosophy and Mathematics to Artificial Intelligence.
 - c. State the relationship between agents and environment.
 - d. What is PEAS description? Explain with two suitable examples.
 - e. Explain following task environments:
 - i) Single Agent vs. Multiagent
 - ii) Episodic vs. Sequential
 - f. Describe the structure of Utility based Agent.

2. Attempt any three of the following: 15
 - a. Describe the problem formulation of Vacuum World problem.
 - b. Explain following terms:

i) State Space of problem	ii) Path in State Space
iii) Goal Test	iv) Path Cost
 - c. Give the outline of Breadth First Search algorithm with respect to Artificial Intelligence.
 - d. With the Local Search algorithm, explain the following concepts:

i) Shoulder	ii) Global Maximum	iii) Local Maximum
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 - e. Illustrate Hill Climbing algorithm using 8 queen problem.
 - f. Explain the mechanism of Genetic Algorithm.

3. Attempt any three of the following: 15
 - a. Explain Minimax algorithm in detail.
 - b. Describe the technique of Alpha-Beta Pruning.
 - c. Write a short note on Kriegspiel's Partially observable chess.
 - d. What is knowledge based agent? Explain its importance in problem solving techniques.
 - e. Write a short note on Wumpus world problem.
 - f. Explain Forward-Chaining algorithm for Propositional definite Clauses.

4. Attempt any three of the following: 15
 - a. What is meant by First Order Logic? Explain syntax and semantics of First Order Logic.
 - b. Write a short note on Universal and Existential quantifier with suitable example.
 - c. Explain the steps of Knowledge Engineering projects in First Order Logic.
 - d. Write a short note on Unification Process.
 - e. Explain Datalog used in first order definite clause.
 - f. Describe Backward-Chaining algorithm for First Order definite Clauses.

[Turn over...]

5. Attempt any three of the following: 15
- a. Explain Planning Domain Definition Language description for an Air Cargo planning problem.
 - b. Describe Forward (Progression) State-Space Search algorithm with an example.
 - c. Explain hierarchical planning.
 - d. Write a short note on Sensorless Planning Problem.
 - e. What are events? Explain its importance.
 - f. What is semantic network? Show the semantic network representation with a suitable example.
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